

Understanding Cross-Domain Robustness in Harmful Meme Detection with Vision-Language Models

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Abstract

Detecting harmful content in internet memes remains a challenging multimodal reasoning problem due to the complex interaction between visual and textual modalities, implicit semantics, contextual ambiguity, and rapidly evolving online discourse. Although recent multimodal learning systems and large vision-language models (LVLMs) have achieved strong benchmark performance, their robustness and transferability under cross-domain settings remain insufficiently understood.

This work presents a robustness-oriented analytical framework for cross-domain harmful meme understanding under semantic distribution shifts. Rather than proposing a new architecture, the study systematically investigates how domain adaptation, contrastive representation alignment, fusion-based multimodal learning, and instruction-aware vision-language models influence robustness, harmful-content sensitivity, and representation transferability across heterogeneous meme domains.

Experiments are conducted using Memotion as the source domain and Hateful Memes as the target domain under low-resource adaptation settings. The results reveal a clear trade-off between overall classification accuracy and harmful-content recall. Domain adaptation improves global classification performance, whereas contrastive alignment enhances sensitivity toward harmful and implicitly offensive content through semantically transferable representations. Furthermore, evaluation of BLIP-2 and LLaVA-1.5 demonstrates that modern LVLMs exhibit substantially different robustness profiles under domain shift. While BLIP-2 achieves the strongest balance between accuracy and harmful-content sensitivity, LLaVA-1.5 attains higher precision but substantially lower harmful recall,

indicating that multimodal reasoning capability alone does not guarantee robust harmful meme understanding. The study further provides analytical insights into multimodal robustness, semantic transferability, and reasoning limitations of modern vision–language models under heterogeneous meme distributions. Overall, the study provides analytical insights into multimodal robustness, semantic transferability, and harmful-content sensitivity under cross-domain settings, highlighting the importance of robustness-aware evaluation for trustworthy multimodal moderation systems operating under evolving online discourse environments.

Keywords: Cross-domain harmful meme detection, foundation models, vision-language models, transferability, multimodal reasoning, domain adaptation, contrastive learning, robustness under distribution shift

1 Introduction

The rapid proliferation of social media platforms has fundamentally transformed the way information, opinions, and social narratives are communicated online. Among the various forms of user-generated content, internet memes have emerged as one of the most influential multimodal communication mechanisms, combining textual and visual elements to express humor, political commentary, emotions, sarcasm, and social perspectives. Due to their compact and highly expressive nature, memes are capable of conveying complex semantic meanings through subtle interactions between image and text. While memes are frequently used for entertainment and social engagement, they are also increasingly exploited to disseminate harmful, hateful, offensive, or misleading content across online platforms.

Unlike traditional text-based abusive language detection, harmful meme understanding presents a substantially more challenging multimodal reasoning problem. In many cases, harmful intent does not explicitly appear in either the textual or visual modality individually, but instead emerges through implicit interactions between modalities. Harmful memes often rely on contextual abstraction, sarcasm, cultural references, symbolic visual cues, or implicit semantic relationships that require deeper multimodal interpretation. Consequently, conventional unimodal moderation approaches frequently fail to capture the nuanced semantics associated with harmful meme content.

Recent advances in multimodal learning and vision–language representation modeling have significantly improved performance across a variety of multimodal understanding tasks. Transformer-based architectures and contrastive multimodal representation learning methods, such as CLIP, have demonstrated strong capabilities in aligning textual and visual semantics within shared embedding spaces [1]. Furthermore, the emergence of large vision–language models (LVLMs), including BLIP-2, MiniGPT-4, and LLaVA, has substantially advanced multimodal reasoning, contextual understanding, and compositional semantic interpretation through stronger integration between visual encoders and large language models [2, 3]. These systems

have shown increasingly promising results in tasks involving image captioning, visual question answering, multimodal reasoning, and contextual semantic interpretation.

Despite these advances, harmful meme understanding remains considerably more difficult than conventional multimodal classification tasks. Recent studies suggest that harmful meme interpretation frequently involves implicit semantics, multimodal abstraction, weak image-text grounding, and rapidly evolving online discourse patterns [4, 5]. In many instances, harmful content is embedded through subtle contextual relationships rather than explicit offensive expressions. As a result, even modern multimodal systems may struggle to correctly interpret harmful intent when contextual reasoning, external knowledge, or semantic disambiguation becomes necessary.

An additional challenge arises from the highly heterogeneous nature of online meme ecosystems. Existing multimodal harmful meme detection systems are commonly evaluated under single-domain settings, where training and testing samples originate from similar semantic distributions. However, real-world meme distributions vary substantially across platforms, communities, cultures, languages, annotation schemes, and evolving online trends. Memes collected from different datasets often differ significantly in visual style, humor patterns, symbolic semantics, linguistic expressions, and contextual framing. Consequently, multimodal models trained on one dataset frequently experience substantial performance degradation when applied to unseen target domains, highlighting the broader problem of *cross-domain robustness* and representation transferability.

Recent research in domain generalization and robust multimodal learning further indicates that modern deep learning systems frequently overfit dataset-specific statistical patterns and shortcut correlations rather than learning semantically transferable representations [6, 7]. Similar limitations have also been observed in multimodal vision-language systems, where weak semantic grounding and spurious multimodal correlations reduce generalization capability under distributional shifts [8]. These challenges become particularly severe in harmful meme detection because harmful semantics often depend on implicit multimodal interactions that evolve dynamically across online environments.

Although prior studies have achieved promising benchmark performance on datasets such as Hateful Memes and Memotion, comparatively limited attention has been given to systematically understanding how modern multimodal learning paradigms behave under cross-domain harmful meme understanding settings. Most existing studies primarily emphasize in-domain benchmark optimization and architectural improvements, while the robustness, transferability, calibration behavior, and harmful-content sensitivity of multimodal systems under heterogeneous distributions remain insufficiently explored.

Motivated by these limitations, this work adopts an analytical perspective to systematically investigate cross-domain harmful meme understanding under domain shift conditions. Rather than proposing a highly optimized benchmark-oriented architecture, the primary objective of this study is to analyze how different multimodal learning paradigms influence robustness, harmful-content sensitivity, and representation transferability across heterogeneous meme distributions. Specifically, we investigate

the complementary effects of data-level domain adaptation, representation-level alignment through contrastive learning, and multimodal fusion strategies within a unified analytical framework.

The experimental analysis is conducted using the Memotion [9] dataset as the source domain and the Hateful Memes [10] dataset as the target domain under low-resource cross-domain evaluation settings. Through this setup, we analyze how multimodal representations behave under semantic distribution shifts and examine the trade-offs between global classification accuracy, harmful-content recall, and cross-domain robustness.

The findings reveal that optimizing solely for overall classification performance does not necessarily improve harmful meme understanding under cross-domain conditions. While adaptation-oriented learning strategies improve global decision-boundary adaptation, representation-level alignment enhances sensitivity toward harmful and implicitly offensive content. The analysis further demonstrates that balancing robustness, transferability, calibration reliability, and harmful-content sensitivity remains a fundamental challenge for multimodal moderation systems operating under evolving online environments.

The main contributions of this work are summarized as follows:

- A robustness-oriented analytical framework for cross-domain harmful meme understanding under semantic distribution shifts is presented.
- The study systematically evaluates both lightweight multimodal learning frameworks and modern instruction-aware vision–language models under heterogeneous meme distributions.
- The work provides analytical insights into the trade-offs between classification accuracy, harmful-content sensitivity, semantic transferability, and contextual robustness.
- The proposed evaluation setting investigates how representation alignment, domain adaptation, and multimodal reasoning influence robustness under cross-domain conditions.
- Qualitative reasoning analysis reveals important multimodal failure modes involving implicit semantics, contextual ambiguity, symbolic stereotypes, sarcasm, and weak multimodal grounding.

2 Related Work

2.1 Harmful Meme Detection

The rapid proliferation of multimodal content on social media platforms has intensified the spread of harmful memes containing hate speech, misinformation, harassment, and offensive narratives. Unlike conventional hate speech detection, harmful meme detection requires joint understanding of visual and textual modalities, as harmful intent often emerges from the semantic interaction between image and text rather than from either modality independently.

The introduction of the Hateful Memes dataset [10] significantly accelerated research in multimodal harmful content detection by demonstrating the limitations

of unimodal systems. Early approaches primarily relied on convolutional neural networks for visual representation learning and transformer-based architectures for textual understanding. However, these methods often struggled to capture complex cross-modal semantic relationships and contextual dependencies required for meme interpretation.

Subsequent advances in multimodal representation learning introduced transformer-based vision–language architectures such as ViLBERT [11], Visual-BERT [12], LXMERT [13], UNITER [14], and CLIP [1], which demonstrated strong performance in multimodal retrieval and classification tasks through aligned image–text embeddings. Recent studies further demonstrated that incorporating auxiliary metadata such as race, gender, web entities, and emotion cues can improve multimodal hateful meme classification performance by enhancing contextual semantic understanding [15]. More recent vision–language frameworks such as BLIP and BLIP-2 further improved multimodal representation learning by integrating large-scale language supervision with image understanding, demonstrating strong generalization capabilities across diverse multimodal tasks [2, 16]. Recent studies further highlighted that memes frequently exhibit noisy OCR text, weak image–text correlation, and implicit semantic structures, making multimodal reasoning substantially more challenging than traditional vision–language tasks [17]. Prior studies further observed that text-only transformers can sometimes outperform multimodal architectures in hateful meme detection, suggesting that effective multimodal grounding and cross-modal reasoning remain challenging under semantically diverse meme distributions [18]. Parallel research on meme understanding has also explored emotional, sarcastic, and offensive meme analysis using benchmark datasets such as Memotion 2 [19], and Memotion 3 [20]. These studies emphasized the growing semantic and linguistic diversity of memes, including multilingual and code-mixed content. Additional datasets such as MultiOFF further emphasized the importance of multimodal offensive content detection under noisy and socially diverse meme distributions [21]. Emotion-aware multimodal frameworks further demonstrated that effective meme interpretation often depends on fine-grained multimodal interactions and contextual semantic reasoning [22].

Recent surveys additionally highlight the rapid growth of toxic meme analysis research and identify key challenges related to multimodal reasoning, explainability, robustness, and domain generalization in harmful meme detection systems [23]. Recent studies have increasingly explored robustness, semantic alignment, contextual reasoning, and large vision–language model integration for harmful meme understanding. The author in [24] investigated robustness limitations of harmful meme detection systems under adversarial and distributional variations, highlighting the vulnerability of multimodal moderation systems under semantic perturbations. Similarly, in [25] author proposed cross-modal alignment enhancement through informative captioning, whereas in [26] author leveraged frozen vision–language models for harmful meme understanding. More recent works further explored chain-of-thought reasoning, multimodal explainability, and multimodal knowledge augmentation for harmful meme detection using modern vision–language systems and large multimodal models [27–30].

These studies collectively demonstrate the growing importance of contextual reasoning, semantic grounding, and transferable multimodal representations for robust harmful meme understanding.

Despite these advances, most existing studies primarily evaluate models in single-domain settings where training and testing samples share similar distributions. Consequently, the generalization ability of multimodal harmful meme detection systems across heterogeneous datasets remains insufficiently explored. Recent large-scale reasoning-oriented meme datasets further suggest that harmful meme understanding increasingly requires contextual and compositional multimodal reasoning beyond simple image-text alignment [28].

2.2 Large Vision–Language Models for Meme Understanding

The rapid advancement of large vision–language models (LVLMs) has substantially transformed multimodal understanding research by enabling stronger contextual reasoning, compositional semantic interpretation, and multimodal instruction following. Unlike earlier multimodal architectures that primarily relied on shallow image-text alignment, modern LVLMs integrate large language models with visual encoders to perform more sophisticated multimodal reasoning across complex semantic tasks.

Recent frameworks such as BLIP-2 [2] introduced efficient multimodal bootstrapping strategies by connecting frozen visual encoders with large language models, enabling improved zero-shot and transfer learning capabilities across diverse vision–language tasks. Similarly, MiniGPT-4 [3] demonstrated that aligning large language models with visual representations can substantially enhance contextual multimodal reasoning and compositional semantic understanding. More recent conversational vision–language systems such as LLaVA-1.5 further improved multimodal instruction tuning, contextual reasoning, and multimodal semantic interpretation through large-scale visual instruction tuning and enhanced multimodal alignment strategies [31].

Recent large multimodal models such as Qwen2-VL additionally demonstrated strong performance in visual reasoning, OCR understanding, contextual grounding, and multimodal instruction-following tasks through unified multimodal representation learning and large-scale pretraining [32]. These advances suggest that modern LVLMs possess substantially stronger multimodal reasoning capabilities compared to earlier embedding-based multimodal systems.

Within harmful meme understanding, recent reasoning-oriented datasets and frameworks further indicate that meme interpretation increasingly requires contextual semantic reasoning beyond simple image-text matching. Large-scale datasets such as MemeMind [28] emphasize chain-of-thought reasoning, implicit semantic interpretation, and compositional multimodal understanding for harmful meme analysis. These studies highlight that harmful meme understanding often depends on contextual ambiguity resolution, symbolic reasoning, cultural references, and implicit multimodal semantics that remain challenging even for modern multimodal systems.

Despite recent progress, comparatively limited attention has been given to systematically understanding the robustness and transferability of LVLMs under cross-domain harmful meme understanding settings. Most existing studies primarily focus on benchmark optimization under in-domain settings, while the behavior of modern

multimodal reasoning systems under heterogeneous meme distributions and semantic distribution shifts remains insufficiently explored.

2.3 Domain Adaptation and Robustness in Multimodal Learning

Domain adaptation and domain generalization have been extensively studied in computer vision and natural language processing to address performance degradation caused by distributional shifts between source and target domains. In multimodal learning, domain adaptation aims to improve transferability by reducing discrepancies between multimodal feature distributions across datasets. Prior studies on abusive language transfer learning further demonstrated that differences in annotation schemes, cultural contexts, and dataset construction can substantially reduce cross-domain generalization performance, emphasizing the challenges associated with learning transferable representations across heterogeneous distributions [33].

Recent studies on multimodal harmful meme detection have highlighted significant robustness limitations under cross-domain settings. Existing multimodal classifiers often remain vulnerable to semantic perturbations, adversarial variations, and dataset-specific biases, resulting in substantial performance degradation under distributional shifts [24]. Cross-domain investigations further reveal that multimodal meme detectors frequently rely on domain-sensitive visual cues, while textual representations tend to exhibit comparatively stronger transferability across datasets [34]. These findings suggest that learning stable multimodal representations under heterogeneous semantic distributions remains an open challenge.

Prior studies further highlighted that harmful memes frequently contain rapidly evolving semantic references, novel visual entities, and contextual cultural cues, making robust cross-domain generalization particularly challenging for multimodal systems [4]. Recent large-scale multimodal reasoning datasets further suggest that harmful meme understanding increasingly requires contextual and compositional reasoning capabilities beyond shallow image-text alignment [28]. Recent large vision-language models and instruction-tuned multimodal systems further demonstrate strong contextual reasoning and multimodal understanding capabilities; however, their robustness under semantic distribution shifts and evolving meme environments remains comparatively underexplored. Existing studies suggest that even modern LVLMs may remain vulnerable to multimodal hallucination, weak semantic grounding, and shortcut correlations when exposed to heterogeneous multimodal distributions [3, 31, 32]. Recent multimodal meme studies further showed that naive augmentation strategies do not necessarily improve generalization performance and may even increase false negative predictions for harmful categories under imbalanced settings [15].

More broadly, recent surveys on domain generalization emphasize that deep neural networks frequently overfit source-domain characteristics and therefore struggle to generalize under unseen distributions [7, 35]. Similar limitations have also been observed in multimodal vision-language systems, where weak multimodal grounding, shortcut learning behaviors, and dataset-specific biases reduce robustness and semantic generalization capabilities [6, 8, 36]. Furthermore, studies on out-of-distribution robustness

demonstrate that achieving stable performance under distributional variations remains a major challenge for modern deep learning systems [37].

To improve robustness and representation transferability under distributional shifts, several recent approaches have explored representation alignment using contrastive learning objectives and multimodal embedding consistency. CLIP-style alignment learning and recent large-scale vision-language frameworks such as BLIP and BLIP-2 have demonstrated strong performance in multimodal tasks by learning semantically aligned image-text representations [1, 2, 16]. Similarly, recent alignment frameworks such as CapAlign [25] showed that improved cross-modal representation consistency can enhance semantic robustness and transferability in multimodal systems.

Nevertheless, comparatively limited attention has been given to systematically analyzing how different adaptation, alignment, and multimodal reasoning strategies influence cross-domain harmful meme understanding under heterogeneous semantic distributions. In particular, the interaction between representation transferability, harmful-content sensitivity, calibration reliability, and robustness under domain shift remains insufficiently understood, especially in low-resource target-domain settings where labeled multimodal supervision is limited.

2.4 Research Gap and Motivation

Although prior studies have achieved promising performance on benchmark datasets, most existing multimodal harmful meme detection systems are primarily evaluated under in-domain settings where training and testing samples originate from similar semantic distributions. Consequently, comparatively limited attention has been given to systematically understanding how modern multimodal systems behave under cross-domain distribution shifts and heterogeneous meme environments.

Recent advances in large vision-language models have substantially improved multimodal reasoning and contextual semantic understanding. However, existing studies continue to focus predominantly on benchmark optimization and architectural improvements while providing comparatively limited analysis of robustness, representation transferability, harmful-content sensitivity, and calibration behavior under evolving multimodal distributions. In particular, the robustness of multimodal representations under implicit semantics, contextual ambiguity, and heterogeneous online meme distributions remains insufficiently understood. Moreover, comparatively limited attention has been given to robustness-oriented analytical evaluation that jointly investigates semantic transferability, harmful-content sensitivity, contextual reasoning, and reliability behavior under cross-domain multimodal settings.

Furthermore, harmful meme understanding presents challenges that extend beyond conventional multimodal classification tasks. Harmful intent frequently depends on subtle multimodal interactions, symbolic references, contextual semantics, and culturally evolving discourse patterns that may vary significantly across domains. As a result, multimodal systems trained on one dataset may fail to generalize effectively when exposed to unseen semantic distributions.

Motivated by these limitations, this work adopts an analytical perspective to systematically investigate cross-domain harmful meme understanding under domain shift

conditions. Rather than proposing a highly optimized benchmark-oriented architecture, the primary objective is to analyze how different multimodal learning paradigms influence robustness, harmful-content sensitivity, and representation transferability across heterogeneous meme distributions.

The study further investigates the trade-offs between overall classification accuracy, harmful meme recall, and multimodal robustness, providing analytical insights into the limitations of current multimodal moderation systems operating under dynamic and evolving online environments.

3 Dataset Description and Analysis

To investigate cross-domain robustness in multimodal harmful meme understanding, we conduct experiments using two widely adopted multimodal meme datasets: Memotion [9] and Hateful Memes [10]. These datasets differ substantially in annotation objectives, semantic composition, contextual characteristics, and multimodal reasoning complexity, making them suitable for evaluating cross-domain transferability under heterogeneous meme distributions.

The Memotion dataset consists of 6,992 meme instances annotated across multiple affective dimensions, including humour, sarcasm, offensiveness, and motivational content. In this study, we utilize the offensive labels to construct a binary harmful meme detection setting, resulting in 4,279 harmful and 2,713 non-harmful samples. The dataset primarily contains memes characterized by explicit emotional expressions, sarcasm, and visually grounded humor patterns.

In contrast, the Hateful Memes dataset is specifically designed to capture implicit harmful intent that emerges through multimodal interactions between image and text. Unlike Memotion, harmful semantics in Hateful Memes frequently depend on contextual ambiguity, symbolic references, implicit stereotypes, and weak image-text correlations, thereby requiring deeper multimodal reasoning capabilities for accurate interpretation.

A central challenge addressed in this work is the substantial *cross-domain gap* between these datasets. While Memotion primarily reflects emotionally expressive and sarcasm-oriented meme distributions, Hateful Memes focuses more heavily on implicit and context-dependent harmful semantics. Consequently, multimodal systems trained on one dataset may fail to generalize effectively when exposed to differing semantic structures, annotation schemes, and multimodal reasoning requirements in another domain.

Such cross-domain discrepancies are further amplified by annotation subjectivity, cultural variability, evolving online discourse, and rapidly changing meme semantics commonly observed in real-world social media environments [5]. Recent meme datasets and multimodal reasoning studies further suggest that harmful meme understanding increasingly involves multilingual expressions, contextual abstraction, symbolic semantics, and compositional multimodal reasoning beyond simple image-text alignment [20, 28].

For cross-domain adaptation, we utilize a low-resource target-domain setting in which 10% of the Hateful Memes samples are incorporated during training, while

the remaining samples are reserved for evaluation. This setup simulates a realistic deployment scenario where only limited labeled target-domain supervision is available.

Furthermore, both datasets exhibit noticeable class imbalance, which may bias multimodal models toward majority-class predictions under cross-domain settings. To mitigate this issue, class-balanced training and weighted sampling strategies are employed during adaptation experiments.

Table 1 summarizes the statistical characteristics of the datasets used in this study.

Table 1 Statistical summary of datasets used for cross-domain harmful meme understanding

Dataset	Samples	Harmful (%)	Non-harmful (%)
Memotion	6992	61.2	38.8
Hateful Memes (subset)	2000	29.5	70.5

The Memotion dataset exhibits a comparatively higher proportion of harmful content, whereas the Hateful Memes subset is predominantly non-harmful. This imbalance reflects differences in dataset construction methodologies and annotation objectives, further emphasizing the challenges associated with cross-domain robustness and harmful-content sensitivity. In particular, the relatively smaller proportion of harmful samples in the Hateful Memes dataset substantially increases the difficulty of detecting minority harmful instances under distributional shifts.

Overall, the selected datasets provide a suitable benchmark for systematically analyzing multimodal robustness, representation transferability, and harmful-content sensitivity under heterogeneous meme distributions and evolving online semantic environments.

4 Methodology

4.1 Problem Definition

We address the task of *cross-domain harmful meme detection*, where a model trained on a source dataset must generalize to a target dataset with different semantic characteristics. Each meme is represented as an image-text pair (I, T) , and the objective is to predict a binary label $y \in \{0, 1\}$, where $y = 1$ denotes harmful content and $y = 0$ denotes non-harmful content.

Unlike conventional supervised learning settings, this task introduces a substantial *domain shift* between the source domain (Memotion) and the target domain (Hateful Memes). These datasets differ considerably in annotation schemes, meme styles, contextual semantics, and the degree of implicit reasoning required for harmful content understanding. Consequently, models trained exclusively on the source domain often fail to generalize effectively under heterogeneous target-domain conditions.

In this work, our objective is not solely to improve classification performance, but rather to systematically analyze how different multimodal learning strategies influence

robustness, harmful-content sensitivity, and representation transferability under cross-domain settings. To facilitate systematic analytical evaluation under computationally feasible conditions, the experiments primarily employ lightweight multimodal representation frameworks while additionally positioning the analysis within the broader context of modern vision–language systems.

4.2 Overall Framework

Figure 1 illustrates the robustness-oriented analytical framework adopted in this study for cross-domain harmful meme understanding. The framework integrates multimodal representation extraction, cross-domain adaptation, representation-level alignment, and fusion-based evaluation strategies within a unified analytical pipeline. Rather than emphasizing benchmark-oriented architectural optimization, the framework is designed to systematically investigate robustness, harmful-content sensitivity, and representation transferability under heterogeneous multimodal distributions and semantic domain shift conditions. The formulation further enables analytical evaluation of how multimodal representations behave under implicit semantics, contextual variability, and evolving online meme environments. The analytical evaluation framework consists

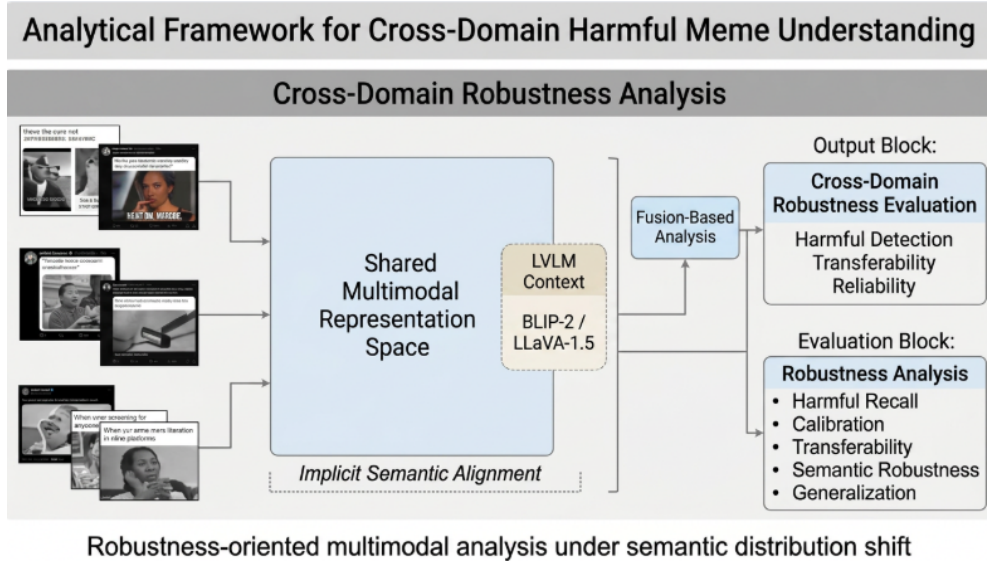


Fig. 1 Overview of the proposed robustness-oriented analytical framework for cross-domain harmful meme understanding. The framework integrates multimodal representation learning, cross-domain adaptation, representation-level alignment, and fusion-based evaluation to systematically analyze robustness, harmful-content sensitivity, and representation transferability under heterogeneous multimodal distributions and semantic domain shifts.

of four major stages: multimodal feature extraction, cross-domain learning strategies, adaptive fusion, and analytical evaluation. Each component is designed to investigate

how different learning paradigms influence robustness and harmful-content sensitivity under cross-domain multimodal settings.

4.3 Multimodal Representation Learning

To capture the joint semantics of visual and textual modalities, we employ a pre-trained CLIP model, which projects images and text into a shared multimodal embedding space. Given an input meme represented as an image-text pair (I, T) , the CLIP encoders generate an image embedding $\mathbf{v}_I$ and a text embedding $\mathbf{v}_T$, respectively.

These embeddings encode high-level semantic relationships between modalities and enable the framework to capture multimodal interactions that are essential for harmful meme understanding. Unlike conventional unimodal classification tasks, harmful intent in memes often emerges from subtle relationships between textual and visual cues rather than from either modality independently. Although the primary experiments employ CLIP-based multimodal representations due to their computational efficiency and strong semantic alignment capabilities, the analytical framework is designed to remain compatible with modern large vision-language models that similarly rely on multimodal representation alignment and contextual semantic grounding. The final multimodal representation is constructed by concatenating the image and text embeddings:

$$\mathbf{x}_m = [\mathbf{v}_I; \mathbf{v}_T] \quad (1)$$

where $\mathbf{x}_m$ denotes the combined multimodal representation used for downstream learning and analytical evaluation.

The resulting embedding space serves as the foundation for investigating how different learning strategies influence robustness, representation transferability, and harmful-content sensitivity under cross-domain settings.

4.4 LVLM-Based Cross-Domain Evaluation

In addition to lightweight multimodal representation frameworks, this study further positions the analytical evaluation within the broader context of modern large vision-language models (LVLMs). Recent LVLMs have demonstrated strong capabilities in multimodal reasoning, contextual semantic interpretation, visual instruction following, and compositional understanding through large-scale multimodal pretraining and alignment with large language models. Consequently, evaluating robustness under cross-domain harmful meme settings requires considering how such systems behave under heterogeneous multimodal distributions and evolving semantic environments.

Within this study, BLIP-2 and LLaVA-1.5 are experimentally evaluated as representative instruction-aware LVLM paradigms because they incorporate multimodal semantic grounding and contextual reasoning mechanisms that extend beyond conventional image-text classification [2, 31]. Unlike earlier multimodal systems that primarily depend on shallow feature alignment, these models integrate visual encoders with large language models to perform instruction-aware multimodal reasoning and contextual semantic interpretation.

Other emerging multimodal reasoning frameworks, including MiniGPT-4 and Qwen2-VL, are also conceptually relevant to the broader evolution of LVLM research [3, 32]. However, these models are discussed as representative examples of recent reasoning-oriented multimodal architectures rather than being included in the experimental evaluation.

The proposed analytical framework is therefore designed to remain compatible with both lightweight representation-learning approaches and modern LVLM-based multimodal reasoning systems. In particular, the study focuses on understanding how multimodal representations behave under semantic distribution shifts, implicit harmful semantics, and heterogeneous meme environments rather than restricting evaluation to a single architectural paradigm.

From an analytical perspective, cross-domain harmful meme understanding presents a particularly challenging setting for LVLMs because harmful intent frequently depends on implicit multimodal interactions, contextual ambiguity, symbolic references, sarcasm, and culturally evolving discourse patterns. Under such conditions, even modern multimodal reasoning systems may remain vulnerable to semantic inconsistencies, weak multimodal grounding, shortcut correlations, and domain-sensitive biases.

Consequently, the proposed evaluation setting enables systematic investigation of robustness, harmful-content sensitivity, representation transferability, and calibration reliability under realistic cross-domain multimodal conditions. This formulation further allows the study to bridge lightweight multimodal representation analysis with emerging multimodal reasoning paradigms in modern vision-language systems.

4.5 Data-Level Domain Adaptation

To mitigate the distributional gap between source and target domains, we employ a data-level domain adaptation strategy by incorporating a limited portion of labeled target-domain samples during training.

Given source-domain data (X_s, Y_s) and target-domain data (X_t, Y_t) , a combined adaptation dataset is constructed as:

$$X_{\text{adapt}} = X_s \cup X_t, \quad Y_{\text{adapt}} = Y_s \cup Y_t \quad (2)$$

To address class imbalance and encourage adaptation toward target-domain characteristics, higher sample weights are assigned to target-domain instances during training. This weighting strategy enables the model to leverage the larger source-domain dataset while simultaneously improving sensitivity to target-domain semantic patterns.

The proposed setup simulates a realistic low-resource adaptation scenario in which only limited labeled target-domain supervision is available. Consequently, the framework enables evaluation of cross-domain transferability under practical deployment conditions where fully labeled target-domain data may be unavailable.

This component primarily establishes an adaptation-oriented robustness baseline, allowing us to analyze how far overall performance can be improved through data-level adaptation alone. However, the strategy does not explicitly enforce semantic

alignment between multimodal representations across domains, motivating the need for representation-level alignment techniques.

4.6 Representation-Level Alignment via Contrastive Learning

Although data-level domain adaptation improves overall classification performance, it does not explicitly ensure that multimodal representations remain semantically consistent across heterogeneous domains. To address this limitation, we incorporate a representation-level alignment strategy using contrastive learning objectives.

Given a batch of multimodal embeddings $\{\mathbf{z}_i\}$, the contrastive objective encourages samples belonging to the same class to remain closer in the embedding space while simultaneously increasing separation between samples from different classes. This process promotes semantically aligned and transferable multimodal representations under cross-domain settings. Such representation-level alignment strategies are conceptually related to recent multimodal contrastive pretraining paradigms adopted in modern vision–language systems, where semantic consistency across modalities plays a critical role in improving robustness and transferability under heterogeneous multimodal environments.

The overall optimization objective is defined as:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{cls}} + (1 - \alpha) \mathcal{L}_{\text{con}} \quad (3)$$

where $\mathcal{L}_{\text{cls}}$ denotes the standard cross-entropy classification loss and $\mathcal{L}_{\text{con}}$ represents the contrastive alignment loss. The parameter α controls the trade-off between classification performance and representation alignment. In our experiments, we empirically set $\alpha = 0.7$ to balance global classification accuracy with semantic consistency across domains.

Unlike conventional classification objectives that primarily optimize decision boundaries, contrastive representation alignment explicitly focuses on preserving multimodal semantic relationships under distributional shifts. Consequently, this component enables systematic analysis of whether semantically aligned embeddings improve harmful-content sensitivity and representation transferability, particularly for memes involving implicit, ambiguous, or context-dependent harmful semantics.

4.7 Hybrid Fusion Strategy

A central observation of this study is that different learning strategies optimize distinct aspects of cross-domain harmful meme detection performance. While data-level adaptation improves overall classification accuracy through target-aware learning, representation-level alignment enhances sensitivity toward harmful content by promoting semantically consistent multimodal embeddings.

To leverage these complementary strengths, we employ a hybrid fusion strategy that combines predictions from the adaptation-based and contrastive learning models. Let P_{adapt} denote the prediction probability from the domain-adapted classifier and P_{con} denote the probability generated by the contrastive alignment model. The final prediction probability is computed as:

$$P_{\text{final}} = \lambda P_{\text{adapt}} + (1 - \lambda) P_{\text{con}} \quad (4)$$

where $\lambda \in [0, 1]$ controls the contribution of each learning component. Based on empirical validation, we set $\lambda = 0.7$ to prioritize classification stability while preserving harmful-content sensitivity.

The fusion strategy serves as an analytical mechanism for examining how complementary adaptation-oriented and representation-oriented learning paradigms influence robustness and harmful-content sensitivity under cross-domain settings. Consequently, the fusion framework enables systematic analysis of how complementary learning paradigms influence robustness, transferability, and harmful-content sensitivity under heterogeneous multimodal distributions.

4.8 Calibration Analysis

To further investigate prediction reliability under cross-domain settings, we analyze model calibration behavior using probability calibration techniques. Calibration analysis evaluates whether predicted confidence scores accurately reflect the true likelihood of model predictions.

This analysis is particularly important in harmful meme detection because semantic ambiguity and distributional shifts can substantially affect confidence reliability even when classification accuracy appears acceptable. Under heterogeneous multimodal conditions, models may exhibit uncertainty due to weak image-text correlations, implicit harmful semantics, or inconsistent domain-specific patterns.

Our experiments show that although calibration improves confidence alignment, it may simultaneously reduce classification performance, highlighting an important trade-off between predictive reliability and classification effectiveness. These observations suggest that achieving both accurate and well-calibrated harmful meme detection under cross-domain settings remains a challenging problem for current multimodal learning systems. These observations further suggest that reliability-aware evaluation remains essential for modern multimodal moderation systems, particularly under evolving semantic environments and heterogeneous online distributions.

4.9 Analytical Perspective

Unlike conventional benchmark-oriented studies that primarily focus on maximizing classification performance, this work adopts an analytical perspective to investigate how different multimodal learning strategies behave under cross-domain harmful meme detection settings. The primary objective is not merely to optimize predictive accuracy, but to systematically analyze the trade-offs between robustness, harmful-content sensitivity, and representation transferability under heterogeneous multimodal distributions.

Specifically, the proposed framework investigates the complementary effects of:

- data-level domain adaptation for improving target-domain decision-boundary adaptation,
- representation-level alignment through contrastive learning for enhancing multimodal semantic consistency across domains, and

- hybrid fusion strategies for balancing complementary adaptation-based and representation-based learning behaviors.

Through this formulation, the framework enables systematic examination of how different learning paradigms influence cross-domain robustness and harmful meme understanding. Consequently, the study contributes a robustness-oriented analytical investigation of multimodal harmful meme understanding under realistic semantic distribution shifts rather than proposing a single highly optimized benchmark-oriented architecture.

5 Results

5.1 Overall Performance Comparison

To evaluate cross-domain generalization, models were trained on the Memotion dataset and evaluated on the Hateful Memes dataset under a domain adaptation setting. Table 2 summarizes the comparative performance of all evaluated approaches.

The experimental results reveal substantial performance variation across different learning strategies, highlighting the challenges associated with multimodal robustness and representation transferability under heterogeneous meme distributions. The baseline model achieves an accuracy of only 0.48, indicating poor cross-domain generalization when representations are learned exclusively from the source domain. This observation confirms the presence of a significant semantic gap between the two datasets, where patterns learned from Memotion fail to transfer effectively to the target domain.

Interestingly, the baseline model exhibits comparatively high harmful-content recall (0.67), suggesting that the model tends to over-predict harmful instances when exposed to unfamiliar target-domain semantics. This behavior indicates that source-domain representations alone are insufficient for learning stable decision boundaries under distributional shifts.

Incorporating domain adaptation substantially improves overall classification accuracy (0.65), demonstrating the effectiveness of utilizing limited target-domain supervision for refining decision boundaries. However, this improvement is accompanied by a noticeable reduction in harmful-content recall (0.38), indicating that optimization toward dominant target-domain patterns may suppress sensitivity to minority harmful instances. This behavior highlights an important limitation of accuracy-oriented optimization in safety-critical moderation tasks.

Conversely, representation-level alignment through contrastive learning improves harmful meme detection performance, achieving the highest harmful-content F1-score (0.50). This observation suggests that semantically aligned multimodal embeddings contribute to more transferable representations under cross-domain settings, particularly for memes containing implicit or context-dependent harmful intent.

The proposed fusion-based framework provides a more balanced trade-off between overall classification accuracy and harmful-content sensitivity by integrating complementary adaptation-based and representation-based learning signals. These findings

collectively indicate that robust cross-domain harmful meme detection requires both distribution-level adaptation and semantically consistent multimodal representations.

While the above experiments primarily analyze lightweight multimodal representation frameworks, recent large vision–language models introduce stronger contextual reasoning and instruction-aware multimodal understanding capabilities. Consequently, additional robustness analysis is required to evaluate whether modern LVLM paradigms improve harmful-content sensitivity and semantic transferability under cross-domain settings.

Table 2 Cross-domain performance comparison of adaptation, contrastive alignment, and fusion-based learning strategies for harmful meme detection.

Method	Accuracy	Precision (H)	Recall (H)	F1 (H)
Baseline	0.48	0.37	0.67	0.48
Domain Adaptation	0.65	0.48	0.38	0.44
Contrastive	0.62	0.47	0.53	0.50
Fusion (0.7/0.3)	0.64	0.50	0.45	0.48
Confidence Fusion	0.63	0.49	0.49	0.49

5.2 Cross-Domain Robustness Analysis of Vision–Language Models

Recent large vision–language models (LVLMs) demonstrate substantially stronger contextual reasoning and multimodal semantic interpretation capabilities compared to conventional embedding-based multimodal systems. Unlike earlier representation-oriented frameworks that primarily rely on shared embedding alignment, instruction-aware LVLMs integrate large language models with multimodal reasoning mechanisms to improve contextual understanding, semantic grounding, and compositional image–text interpretation.

To investigate whether such reasoning-aware architectures improve robustness under cross-domain harmful meme understanding settings, we additionally evaluate two representative instruction-aware LVLMs, namely BLIP-2 and LLaVA-1.5. Table 3 and Figure 2 summarize the comparative robustness behavior across lightweight multimodal frameworks and modern vision–language systems.

The experimental observations reveal substantial differences in robustness characteristics across multimodal learning paradigms. Conventional embedding-based CLIP representations exhibit relatively high harmful-content recall (0.67) but poor overall accuracy (0.48), indicating unstable decision boundaries and excessive sensitivity toward harmful predictions under semantic uncertainty. Although data-level domain adaptation substantially improves global classification accuracy (0.65), it simultaneously reduces harmful-content recall (0.38), suggesting that adaptation-oriented optimization may suppress sensitivity toward minority harmful instances.

Conversely, representation-level alignment through contrastive learning improves harmful-content sensitivity (0.53) while maintaining comparatively stable overall performance. This behavior indicates that semantically aligned multimodal embeddings contribute more effectively to transferable harmful-content representations under heterogeneous multimodal distributions.

Among the evaluated vision-language models, BLIP-2 achieves the strongest balance between overall classification accuracy and harmful-content sensitivity, obtaining an accuracy of 0.68 and a harmful-content recall of 0.61. These results suggest that instruction-aware multimodal grounding and contextual semantic reasoning substantially improve semantic transferability under domain shift conditions.

Interestingly, LLaVA-1.5 exhibits a different robustness profile. Although the model achieves the highest harmful-content precision (0.63), its harmful-content recall drops to 0.34, indicating a comparatively conservative prediction behavior that frequently misses harmful instances. This observation suggests that advanced instruction tuning and multimodal reasoning capabilities do not necessarily guarantee robust harmful-content sensitivity under cross-domain settings. Rather, the effectiveness of LVLMs appears to depend not only on reasoning capability but also on the transferability of multimodal representations across heterogeneous meme distributions.

Overall, the comparative analysis reveals that different multimodal paradigms optimize distinct robustness properties under semantic distribution shifts. While adaptation-oriented approaches improve global classification accuracy and representation-alignment approaches improve semantic consistency, instruction-aware LVLMs provide stronger contextual understanding and multimodal reasoning. Nevertheless, even modern LVLMs continue to exhibit limitations when confronted with implicit semantics, sarcasm, symbolic stereotypes, and culturally contextualized meme distributions. Consequently, achieving fully robust cross-domain harmful meme understanding remains an open challenge for current multimodal learning frameworks.

Table 3 Cross-domain robustness comparison across multimodal systems under heterogeneous meme distributions.

Model	Accuracy	Precision (H)	Recall (H)	F1 (H)
CLIP Baseline	0.48	0.37	0.67	0.48
Domain Adaptation	0.65	0.48	0.38	0.44
Contrastive CLIP	0.62	0.47	0.53	0.50
Fusion Strategy	0.64	0.50	0.45	0.48
BLIP-2	0.68	0.52	0.61	0.56
LLaVA-1.5	0.57	0.63	0.34	0.44

Figure 2 presents the trade-off between overall classification accuracy and harmful-content recall across the evaluated multimodal systems. The results reveal substantial differences in robustness characteristics under cross-domain conditions. While the CLIP baseline achieves the highest harmful recall, it suffers from poor overall accuracy due to excessive harmful-content predictions. Domain adaptation improves

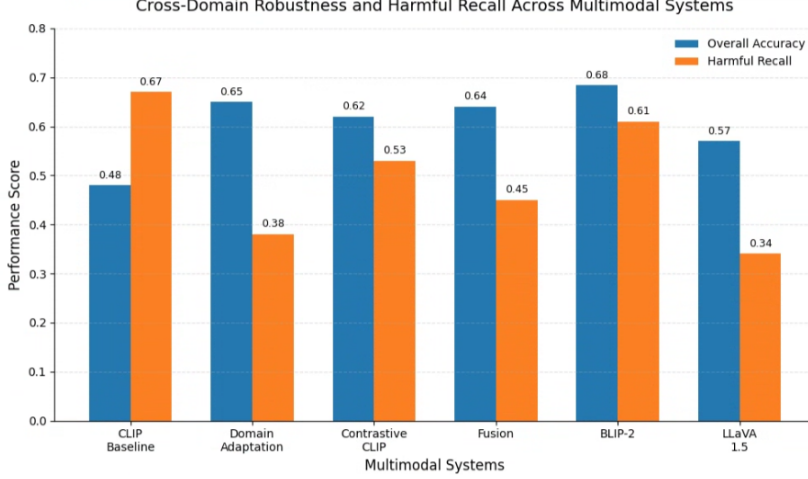


Fig. 2 Cross-domain robustness comparison across multimodal systems. While BLIP-2 achieves the strongest balance between overall accuracy and harmful-content recall, LLaVA-1.5 exhibits a more conservative prediction behavior characterized by higher harmful-content precision but substantially lower harmful recall under domain shift conditions.

overall accuracy but substantially reduces harmful-content sensitivity. In contrast, representation-level alignment through contrastive learning achieves a more balanced trade-off between transferability and harmful-content detection. Among the evaluated vision-language models, BLIP-2 demonstrates the strongest overall robustness, achieving both high accuracy and strong harmful-content recall. LLaVA-1.5 exhibits comparatively lower recall despite instruction-aware multimodal reasoning capabilities, indicating that advanced reasoning mechanisms alone do not necessarily guarantee robust harmful meme understanding under domain shift.

5.3 Robustness and Reliability Evaluation

Although overall classification accuracy is commonly used to evaluate multimodal harmful meme detection systems, accuracy alone is often insufficient for assessing robustness under cross-domain settings. In realistic moderation environments, multimodal systems must simultaneously maintain stable classification performance, sensitivity toward harmful content, and reliable prediction behavior under semantic distribution shifts. Consequently, evaluating cross-domain harmful meme understanding requires a broader robustness-oriented analytical perspective beyond conventional benchmark optimization.

Unlike traditional evaluation protocols that prioritize a single performance criterion, the proposed analytical framework jointly considers classification capability, harmful-content sensitivity, semantic transferability, and prediction reliability under heterogeneous multimodal environments. This perspective is particularly important for harmful meme understanding because moderation systems operating under evolving online distributions must remain sensitive to harmful content while simultaneously avoiding unstable or overconfident predictions.

To support robustness-oriented evaluation, the study incorporates reliability analysis through Brier score estimation for probabilistic multimodal classifiers. Lower Brier scores indicate better calibration quality and more reliable probabilistic behavior under cross-domain conditions. In addition to quantitative reliability analysis, the framework further considers semantic transferability, contextual robustness, and reasoning stability across lightweight multimodal systems and modern instruction-aware vision–language models.

Table 4 summarizes the robustness-oriented comparative evaluation across all multimodal learning paradigms investigated in this study.

Table 4 Robustness-oriented comparison across multimodal systems under cross-domain harmful meme understanding settings.

Model	Accuracy	Recall (H)	Robustness Analysis
CLIP Baseline	0.48	0.67	Unstable decision boundaries
Domain Adaptation	0.65	0.38	Brier Score: 0.2299
Contrastive CLIP	0.62	0.53	Improved semantic transferability
Fusion Strategy	0.64	0.45	Brier Score: 0.3295
BLIP-2	0.68	0.61	Strong contextual robustness
LLaVA-1.5	0.57	0.34	Reasoning instability under shift

The robustness-oriented evaluation reveals substantial differences in reliability and transferability across multimodal learning paradigms. Although certain lightweight models achieve comparatively high harmful-content recall, their unstable calibration behavior and weak cross-domain generalization reduce overall robustness. In contrast, BLIP-2 demonstrates a comparatively stronger balance between classification performance, harmful-content sensitivity, and contextual semantic robustness under semantic distribution shifts.

Interestingly, LLaVA-1.5 exhibits comparatively weaker robustness despite strong multimodal reasoning capabilities and higher harmful-content precision. This observation suggests that instruction-aware reasoning alone may not guarantee stable harmful meme understanding under cross-domain conditions. Rather, robust multimodal moderation appears to depend on a combination of contextual reasoning, semantically transferable representations, calibration reliability, and stable multimodal grounding under heterogeneous semantic environments.

Overall, the proposed robustness-oriented evaluation perspective provides a unified analytical framework for systematically investigating multimodal harmful meme understanding beyond conventional benchmark accuracy. The findings further highlight that balanced evaluation of accuracy, harmful-content sensitivity, reliability, and contextual robustness is essential for designing trustworthy multimodal moderation systems operating under evolving online discourse conditions.

5.4 Trade-off Between Accuracy and Harmful Meme Detection

One of the most important observations in this study is the consistent trade-off between overall classification accuracy and harmful-content sensitivity under cross-domain settings. Figure 3 illustrates the relationship between overall accuracy and harmful meme recall across different learning strategies.

The experimental results indicate that models optimized primarily for global classification accuracy tend to sacrifice sensitivity toward harmful content. In particular, the domain adaptation model achieves the highest overall accuracy (0.65) but exhibits substantially lower harmful-content recall (0.38). This behavior suggests that accuracy-oriented optimization may favor dominant semantic patterns within the target distribution while suppressing sensitivity toward minority harmful instances.

Conversely, representation-level alignment through contrastive learning achieves comparatively stronger harmful-content recall (0.53), despite slightly lower overall accuracy. This observation indicates that semantically aligned multimodal embeddings contribute more effectively to detecting implicit or context-dependent harmful semantics that are difficult to capture through conventional classification objectives alone.

The proposed fusion-based framework provides a more balanced compromise between these competing objectives by integrating both adaptation-based and representation-based learning signals. Although the fusion model does not maximize either accuracy or recall individually, it achieves more stable cross-domain behavior by balancing robustness and harmful-content sensitivity. Interestingly, the instruction-aware BLIP-2 framework achieves a comparatively stronger balance between these competing objectives, suggesting that contextual multimodal reasoning contributes to more stable robustness behavior under semantic distribution shifts.

These findings have important implications for real-world moderation systems. In safety-critical environments, optimizing solely for overall classification accuracy may lead to under-detection of harmful content, particularly under heterogeneous and evolving meme distributions. Consequently, robust harmful meme detection systems should jointly consider representation transferability, semantic robustness, and harmful-content sensitivity rather than relying exclusively on global accuracy optimization. The observed trade-off further highlights the difficulty of maintaining stable harmful-content sensitivity under heterogeneous multimodal distributions.

5.5 Confusion Matrix Analysis

The confusion matrix shown in Figure 4 provides additional insight into the prediction behavior of the fusion-based framework under cross-domain evaluation settings. Although the model successfully identifies a considerable number of harmful instances, it also produces a relatively high number of false positives for benign memes.

This behavior suggests that the model adopts a conservative decision strategy by prioritizing harmful-content detection under semantic uncertainty. Such behavior is commonly observed in safety-critical classification systems, where minimizing false negatives for harmful categories is often prioritized despite increased false-positive rates.

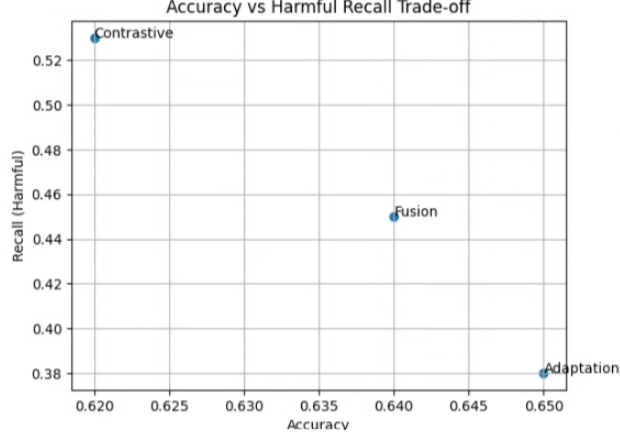


Fig. 3 Relationship between overall classification accuracy and harmful-content recall across different cross-domain learning strategies.

The observed false-positive tendency further indicates that cross-domain distributional shifts introduce ambiguity in multimodal representations, making it difficult for the model to distinguish between subtle semantic variations across datasets. In particular, differences in meme style, contextual framing, and image-text relationships between Memotion and Hateful Memes contribute to unstable decision boundaries under target-domain conditions.

From a practical perspective, this behavior reflects an important trade-off in real-world moderation systems. While conservative prediction strategies may improve harmful-content sensitivity, they can simultaneously increase the misclassification of benign content, thereby affecting moderation reliability and user trust.

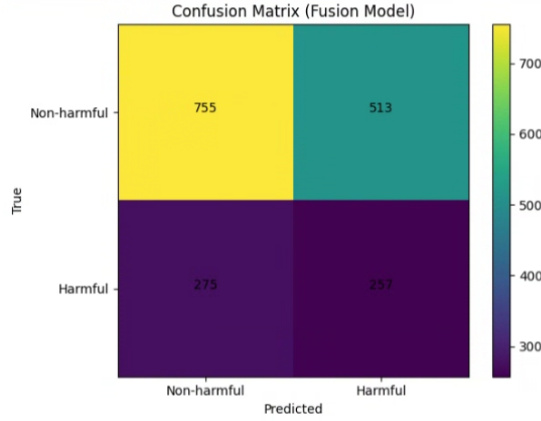


Fig. 4 Confusion matrix illustrating prediction behavior of the fusion-based framework under cross-domain harmful meme detection settings.

5.6 Precision–Recall Analysis

The precision–recall curve shown in Figure 5 further highlights the difficulty of harmful meme detection under cross-domain settings. Precision remains relatively low across different recall levels, indicating that the model struggles to maintain stable confidence when distinguishing harmful from non-harmful instances across heterogeneous multimodal distributions.

This behavior can be attributed to the inherently ambiguous and context-dependent nature of harmful memes, where semantic meaning often emerges from subtle interactions between textual and visual modalities. Unlike explicit offensive content, many harmful memes rely on indirect references, sarcasm, weak image–text correlations, or culturally contextualized semantics that are difficult to capture using conventional multimodal classification objectives.

The observed precision–recall characteristics further suggest that cross-domain harmful meme detection remains highly sensitive to representation inconsistency and semantic variability across datasets. As a result, multimodal models may successfully identify certain harmful patterns while simultaneously exhibiting uncertainty in separating benign and harmful content under distributional shifts.

These findings reinforce the importance of semantically aligned and transferable multimodal representations for improving robustness and confidence reliability in real-world harmful meme moderation systems.

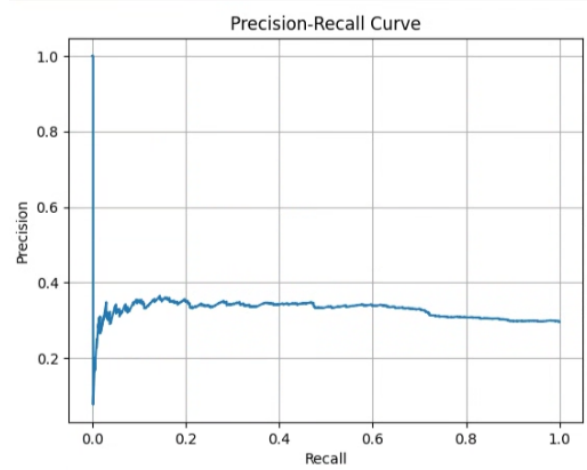


Fig. 5 Precision–recall characteristics of the fusion-based framework under cross-domain harmful meme detection settings.

5.7 Calibration Analysis

The calibration curve shown in Figure 6 provides additional insight into the confidence reliability of the fusion-based framework under cross-domain evaluation settings. The model exhibits a generally under-confident prediction behavior, where predicted

probabilities remain below the ideal calibration diagonal across multiple confidence intervals.

This observation suggests that the model captures relevant multimodal patterns but remains uncertain when assigning confidence scores under heterogeneous target-domain conditions. Such behavior is expected under cross-domain settings, where differences in meme semantics, linguistic structure, visual style, and contextual framing reduce representation consistency and prediction certainty.

The observed calibration characteristics further indicate that domain shift affects not only classification performance but also confidence reliability in multimodal prediction systems. In particular, semantically ambiguous memes and weak image–text correlations introduce uncertainty in the learned representation space, leading to conservative probability estimation.

Interestingly, attempts to improve calibration through post-processing techniques resulted in a reduction in overall classification performance, highlighting an important trade-off between confidence reliability and predictive accuracy. This finding suggests that achieving both accurate and well-calibrated multimodal harmful meme detection under domain shift remains a challenging problem for current multimodal learning systems.

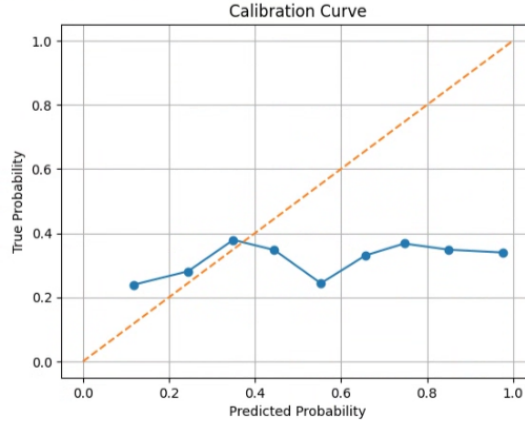


Fig. 6 Calibration characteristics of the fusion-based framework under cross-domain harmful meme detection settings.

5.8 Effect of Representation Alignment and Fusion

The experimental observations indicate that representation-level alignment contributes more strongly to harmful-content sensitivity than to overall classification accuracy. In particular, contrastive learning improves harmful meme detection performance by encouraging semantically consistent multimodal embeddings across

heterogeneous domains. This behavior is especially beneficial for memes containing implicit, context-dependent, or weakly correlated image–text semantics that are difficult to capture using conventional classification objectives alone.

The results further suggest that representation alignment enhances the transferability of multimodal embeddings by reducing semantic inconsistency between source and target distributions. Consequently, contrastive learning improves the detection of subtle harmful relationships that may remain ambiguous under purely adaptation-based learning strategies.

However, representation alignment alone remains insufficient for achieving stable global decision boundaries under cross-domain settings. Although contrastive learning improves harmful-content recall and representation transferability, the absence of stronger target-aware adaptation limits its overall classification performance under heterogeneous multimodal distributions.

The fusion-based framework demonstrates that combining data-level adaptation with representation-level alignment produces more balanced and robust cross-domain behavior. By integrating complementary learning signals, the fusion strategy simultaneously improves decision-boundary adaptation and multimodal semantic consistency. These findings suggest that robust harmful meme detection may require both distribution-level adaptation and transferable multimodal representations rather than relying exclusively on a single learning paradigm.

5.9 Reasoning Failure Analysis of Vision–Language Models

Qualitative analysis of BLIP-2 and LLaVA-1.5 predictions reveals several important robustness limitations under cross-domain harmful meme understanding settings. Although instruction-aware vision–language models demonstrate improved contextual reasoning and semantic interpretation capabilities compared to conventional embedding-based multimodal systems, multiple challenging failure modes remain under heterogeneous meme distributions.

One common failure category involves implicit harmful semantics and culturally contextualized offensive references. Several memes containing symbolic stereotypes, political references, discriminatory humor, or socially contextual harmful implications were incorrectly classified despite containing offensive intent. Representative examples include memes involving racial stereotypes, geopolitical references, and culturally specific offensive expressions, indicating that contextual harmful semantics often remain difficult to interpret under cross-domain conditions where meme structures, humor styles, and semantic framing vary substantially across datasets.

The analysis further suggests that modern LVLMs may occasionally over-rely on surface-level textual semantics while insufficiently grounding multimodal contextual relationships. Certain memes containing provocative or emotionally charged lexical cues were incorrectly predicted as harmful despite lacking explicit abusive intent, indicating sensitivity toward isolated textual patterns rather than holistic multimodal reasoning. Conversely, several harmful memes were incorrectly classified as non-harmful because the offensive meaning emerged only through subtle image–text interactions that required deeper contextual understanding.

Additional failure cases emerged in memes involving sarcasm, implicit discriminatory humor, cultural references, weak OCR grounding, and contextual semantic ambiguity. For example, LLaVA-1.5 frequently failed to identify harmful content embedded within indirect stereotypes and contextual jokes, while BLIP-2 occasionally struggled with highly implicit offensive semantics requiring external contextual knowledge. These observations suggest that current vision-language models still face challenges in integrating symbolic reasoning, contextual grounding, and culturally aware semantic interpretation.

The observed reasoning limitations further highlight that cross-domain harmful meme understanding remains fundamentally different from standard multimodal classification tasks. Unlike explicit offensive language detection, harmful meme understanding often depends on subtle semantic interactions, implicit contextual references, symbolic associations, and evolving online discourse patterns that remain difficult to model using current multimodal systems.

Interestingly, the comparative evaluation also reveals that different LVLMS exhibit distinct robustness profiles. While BLIP-2 achieves stronger harmful-content sensitivity and a more favorable balance between accuracy and recall, LLaVA-1.5 demonstrates higher harmful-content precision but substantially lower harmful recall, indicating a more conservative prediction behavior. This finding suggests that advanced instruction tuning and multimodal reasoning capabilities alone do not necessarily guarantee robust harmful-content understanding under domain shift conditions.

These observations collectively indicate that although instruction-aware multimodal systems substantially improve contextual reasoning and semantic grounding, achieving robust cross-domain harmful meme understanding remains an open challenge due to implicit semantics, cultural variability, contextual ambiguity, weak multimodal grounding, and heterogeneous semantic distributions. Future research may therefore benefit from integrating stronger reasoning-aware architectures, external knowledge grounding, and domain-generalizable multimodal representations to improve robustness under evolving online discourse environments.

Table 5 Representative reasoning failures observed in instruction-aware vision-language models under cross-domain harmful meme understanding.

Meme Text	Ground Truth	Prediction	Failure Type
all races are equal	Harmful	Non-harmful	Implicit harmful semantics ambiguity
when the whole squad roasts the jewish kid	Non-harmful	Harmful	Contextual lexical bias
celebrating the ebt card. oh snap	Harmful	Non-harmful	Cultural stereotype reasoning failure
the enemy within ilhan omar...	Non-harmful	Harmful	Political semantic oversensitivity
when each letter is a mental disorder	Non-harmful	Harmful	Over-reliance on offensive lexical cues

Several failure cases further demonstrate that harmful intent frequently emerges from implicit multimodal context rather than isolated textual content alone. For example, certain memes containing seemingly neutral textual statements were incorrectly classified because the harmful implication was embedded within visual stereotypes, sarcastic framing, or culturally contextual multimodal associations that require deeper semantic grounding beyond surface-level text interpretation.

6 Discussion

6.1 Cross-Domain Performance Analysis

Table 2 summarizes the performance of the evaluated models under the cross-domain setting, where Memotion is used as the source domain and Hateful Memes as the target domain.

The baseline model trained exclusively on the source domain exhibits noticeable performance degradation on the target dataset, highlighting the severity of the semantic and distributional gap between meme domains. This observation is consistent with prior studies on domain generalization, which show that deep learning models often overfit source-domain characteristics and struggle to maintain robustness under unseen distributions [7, 35].

Incorporating a small subset of target-domain samples through domain adaptation improves overall classification accuracy by refining global decision boundaries. However, this improvement is accompanied by comparatively lower harmful-content recall, suggesting that optimization toward dominant target-domain patterns may reduce sensitivity to minority harmful instances.

Conversely, representation-level alignment through contrastive learning improves harmful meme recall by encouraging semantically consistent multimodal embeddings across domains. These findings indicate that contrastive objectives contribute to more transferable representations, particularly for memes containing implicit or context-dependent harmful intent. Similar observations have been reported in multimodal alignment studies, where stronger representation consistency improves semantic robustness under distributional variations [25]. Recent large vision-language systems further suggest that stronger multimodal alignment can enhance contextual understanding and compositional reasoning capabilities [3].

The fusion-based framework achieves a more balanced trade-off between overall classification performance and harmful-content sensitivity by combining the complementary strengths of adaptation-based and representation-based learning. This suggests that robust cross-domain harmful meme detection may require both global distribution adaptation and semantically aligned multimodal representations.

6.2 Robustness and Representation Sensitivity

The experimental results further reveal important robustness limitations of multimodal harmful meme detection systems under cross-domain settings. Although multimodal architectures perform well within individual datasets, their generalization

ability decreases considerably when evaluated across heterogeneous meme distributions. This observation aligns with studies showing that multimodal systems often rely on shortcut correlations and dataset-specific biases rather than robust semantic understanding [6, 8].

Recent cross-domain investigations similarly report that multimodal meme detectors are particularly sensitive to domain-specific visual patterns, whereas textual representations often exhibit comparatively stronger transferability [34]. This may partially explain why contrastive representation alignment improves harmful-content recall in our experiments by encouraging more stable multimodal embeddings across domains.

Our findings also align with prior benchmarking studies in which text-only transformers occasionally outperformed multimodal architectures under semantically diverse meme distributions [18]. Such observations indicate that effective multimodal grounding remains a major challenge, especially when harmful intent depends on subtle contextual interactions rather than explicit textual or visual cues. Existing vision–language transformers similarly struggle with meme understanding because harmful semantics often differ substantially from conventional vision–language tasks such as image captioning or visual question answering [4].

Furthermore, robustness studies on out-of-distribution generalization emphasize that maintaining stable performance under evolving semantic distributions remains a major open challenge for modern deep learning systems [37]. These limitations become particularly critical in harmful meme detection, where meme semantics rapidly evolve across social, cultural, and platform-specific contexts [5].

6.3 Implications of Instruction-Aware Vision–Language Models

Recent instruction-aware vision–language models provide additional insights into cross-domain harmful meme understanding. Compared to conventional embedding-based multimodal systems, both BLIP-2 and LLaVA-1.5 demonstrate the ability to perform contextual multimodal reasoning through large-scale vision–language alignment and instruction tuning. However, the experimental results reveal that modern LVLMs exhibit substantially different robustness characteristics under domain shift conditions.

Among the evaluated LVLMs, BLIP-2 achieves the strongest balance between overall classification performance and harmful-content sensitivity, obtaining an accuracy of 68.4% and a harmful-content recall of 61%. These results suggest that contextual multimodal grounding contributes positively to semantic transferability across heterogeneous meme distributions. In contrast, LLaVA-1.5 achieves the highest harmful-content precision (63%) but substantially lower harmful-content recall (34%), indicating a more conservative prediction behavior that frequently misses harmful instances.

This observation is particularly important because it suggests that advanced multimodal reasoning capabilities alone do not necessarily guarantee robust harmful-content detection under cross-domain conditions. Rather, the effectiveness of modern LVLMs

appears to depend on both contextual reasoning ability and the transferability of multimodal representations across heterogeneous semantic distributions.

Furthermore, the qualitative failure analysis indicates that even modern instruction-aware multimodal systems remain vulnerable to implicit harmful semantics, sarcasm, symbolic stereotypes, weak OCR grounding, and culturally contextualized meme references. These findings suggest that robust harmful meme understanding fundamentally requires deeper contextual reasoning and semantic grounding beyond conventional multimodal alignment. Recent multimodal reasoning and alignment-based approaches further highlight the growing importance of semantically transferable multimodal representations for robust harmful meme understanding under evolving online discourse conditions [25, 28, 30].

Overall, the results indicate that future multimodal moderation systems may benefit from integrating robust domain adaptation strategies with reasoning-aware vision–language architectures to jointly improve semantic transferability, contextual understanding, and harmful-content sensitivity under evolving online discourse conditions.

6.4 Trade-off Between Accuracy and Harmful Meme Recall

One of the most important observations in this study is the consistent trade-off between overall classification accuracy and harmful-content sensitivity. While domain adaptation improves global classification accuracy, it does not necessarily maximize harmful meme recall, particularly under imbalanced and heterogeneous data distributions [15, 37]. In contrast, representation-level alignment improves harmful-content detection but may reduce overall classification performance.

This trade-off has important implications for real-world moderation systems. These observations highlight an important practical challenge for real-world moderation systems: optimizing overall classification accuracy alone may be insufficient if harmful content remains undetected, whereas aggressively maximizing harmful-content sensitivity may increase false positives and reduce system reliability. Conversely, aggressively optimizing harmful-content recall may increase false positives and reduce system reliability, reflecting broader trade-offs commonly observed in robust out-of-distribution learning systems [7, 37].

The proposed fusion framework provides a practical compromise by integrating complementary properties of adaptation-based and contrastive learning strategies. These findings suggest that future multimodal moderation systems may benefit from hybrid learning paradigms that jointly consider robustness, transferability, and harmful-content sensitivity. Table 6 summarizes the behavioral characteristics of different learning strategies under cross-domain multimodal meme detection settings. The comparison highlights that different approaches exhibit distinct trade-offs between classification accuracy, harmful-content sensitivity, and representation transferability.

6.5 Qualitative Observations and Limitations

Qualitative analysis indicates that harmful meme detection becomes particularly challenging when harmful intent is implicit, sarcastic, culturally contextualized, or

Table 6 Comparative robustness characteristics of multimodal learning strategies under cross-domain harmful meme detection settings.

Learning Strategy	Strat-	Primary Strength	Observed Limitation	Cross-Domain Behavior
Baseline Classification		Simple and computationally efficient learning framework	Poor robustness under distributional shifts and limited harmful-content sensitivity	Strong dependence on source-domain characteristics
Domain Adaptation		Improves global classification accuracy through target-aware learning	Reduced harmful meme recall under imbalanced target distributions	Better adaptation to target-domain decision boundaries but limited semantic transferability
Contrastive Representation Alignment		Enhances semantically consistent multimodal embeddings and improves harmful-content recall	Slight reduction in overall classification accuracy	More stable representation transferability across heterogeneous domains
Fusion-Based Framework		Balances complementary strengths of adaptation and representation alignment	Moderate increase in framework complexity	Improved trade-off between robustness, transferability, and harmful-content sensitivity
Instruction-Aware LVLMs (BLIP-2, LLaVA-1.5)		Strong contextual reasoning and multimodal semantic grounding	Performance varies substantially across reasoning architectures and semantic distributions	Improved contextual transferability but still vulnerable to implicit harmful semantics

weakly correlated across modalities. Memes containing indirect references or subtle multimodal inconsistencies frequently remain difficult for both adaptation-based and contrastive models, as effective meme interpretation often requires contextual semantic reasoning beyond shallow image–text alignment [4, 5, 17].

The experiments were conducted using lightweight CLIP-based representations and limited target-domain supervision to maintain computational feasibility. This lightweight experimental setup enables systematic analysis of robustness, transferability, and semantic representation behavior under computationally feasible cross-domain settings. Nevertheless, larger instruction-aware multimodal architectures and stronger target-domain supervision may further improve contextual reasoning and semantic grounding capabilities.

Nevertheless, the primary objective of this work is not to propose a highly complex state-of-the-art architecture, but rather to systematically analyze how different multimodal learning strategies behave under cross-domain harmful meme detection settings. The presented findings provide practical insights into the limitations and

trade-offs associated with multimodal robustness and representation transferability under domain shift conditions.

7 Conclusion

This work presented a robustness-oriented analytical investigation of cross-domain harmful meme understanding under heterogeneous multimodal distributions and semantic domain shifts. Rather than focusing exclusively on benchmark-driven architectural optimization, the primary objective of this study was to systematically analyze how different multimodal learning paradigms influence robustness, semantic transferability, contextual reasoning, and harmful-content sensitivity under realistic cross-domain conditions.

Using Memotion as the source domain and Hateful Memes as the target domain, the study examined adaptation-oriented learning, representation-level alignment through contrastive objectives, fusion-based multimodal learning, and modern instruction-aware vision-language models within a unified analytical framework. The experimental findings demonstrate that different multimodal paradigms optimize fundamentally distinct robustness characteristics. Domain adaptation improves global classification performance through target-aware decision-boundary refinement, whereas representation-level alignment enhances semantic consistency and harmful-content sensitivity under distributional shifts. The results further show that robust harmful meme understanding benefits from balancing adaptation, representation transferability, and contextual multimodal reasoning rather than relying on a single learning strategy.

A key finding of this study is that modern instruction-aware vision-language models exhibit substantially different robustness profiles under cross-domain settings. While BLIP-2 achieves the strongest balance between overall classification accuracy and harmful-content sensitivity, LLaVA-1.5 demonstrates higher harmful-content precision but substantially lower harmful-content recall. These observations suggest that advanced multimodal reasoning capability alone does not necessarily guarantee robust harmful meme understanding under semantic distribution shifts. Instead, the transferability of multimodal representations appears to play a critical role in determining robustness across heterogeneous meme distributions.

Beyond quantitative evaluation, the study further reveals that harmful meme understanding remains highly sensitive to implicit semantics, sarcasm, contextual ambiguity, symbolic stereotypes, cultural variability, and weak image-text grounding. The qualitative reasoning analysis demonstrates that even modern instruction-aware multimodal systems continue to struggle with contextual semantic interpretation under evolving online discourse environments. These observations suggest that cross-domain harmful meme understanding fundamentally differs from conventional multimodal classification tasks and requires deeper contextual reasoning, semantic grounding, and transferable multimodal representations.

More importantly, this work emphasizes that robustness-oriented analytical evaluation is equally important as performance optimization in multimodal harmful meme research. The findings highlight that understanding the trade-offs between classification accuracy, harmful-content recall, semantic robustness, calibration reliability,

contextual reasoning, and representation transferability is essential for designing trustworthy moderation systems operating under dynamic and safety-critical online environments.

Overall, the presented study contributes a robustness-centered and analytical perspective on multimodal harmful meme understanding under semantic distribution shifts. By jointly analyzing adaptation-based learning, representation alignment, fusion strategies, and modern vision-language models, the study provides analytical insights into multimodal robustness, semantic transferability, and reasoning limitations of current multimodal systems under heterogeneous meme distributions. The findings further highlight the importance of transferable semantic representations, contextual multimodal reasoning, and robustness-oriented evaluation for future multimodal moderation frameworks.

As future work, extending the analysis to emerging multimodal large language models, reasoning-aware vision-language systems, domain generalization techniques, calibration-aware multimodal learning, and culturally adaptive multimodal frameworks may further improve semantic transferability, contextual understanding, and robustness under evolving online discourse conditions.

Declarations

- **Funding** There is no funding available.
- **Conflict of interest/Competing interests** There is no conflict of interests in this study.
- **Data availability** The reference of the data supporting the findings of this study are available within the article.
- **Author contribution** Sanower Alam: Conceptualization, Literature survey, Coding and experimentation, Writing—original draft preparation and methodology. Dr Jahiruddin: Supervision, Data visualization, Review and editing.
- **Declaration of AI-Assisted Writing** The authors used AI-based language assistance tools solely for improving clarity, grammar, and readability of the manuscript. All technical content, analysis, and conclusions were developed and verified by the authors.

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